

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)

SEM - I University Examination, January- 2011

M.Tech. (CAD/CAM)

ME – 703 Advanced Dynamics of Machines

Date: 05.01.2011, Wednesday

Time: 10:30 a.m. to 01:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

- Q1 (a)** Discuss on balancing of resultant primary and secondary forces for V-Engine. What is the value of resultant primary force and secondary force for 60° V-Engine in terms mass, angular speed, crank radius and angle θ . 10

OR

- (a) Investigate balancing of two stroke six cylinders diesel engine for primary and secondary forces and couples for reciprocating masses. The firing order is I-V-III-VI-II-IV. 10
- (b) Explain the Active balancing and passive balancing. 02

- Q2** An engine mechanism, the crank $CB = 100$ mm and the connecting rod $BA = 300$ mm with centre of gravity G , 100 mm from B and $\angle BCA = (MM * 10)^\circ$. In the position given, the crankshaft has a speed of 75 rad/s and an angular acceleration of 1200 rad/s^2 . Find 1. Velocity of G and 2. Acceleration of G and angular acceleration of AB by using analytical method. Here MM is your Birth Month. 12

OR

- Q2** Fig.1 shows an quick return mechanism. The driving crank OA is 150 mm long and rotates at a uniform speed of $MM * 50$ r.p.m. in counter clockwise direction. For the position shown in fig.1 Find i) velocity of ram R ii) Acceleration of ram R and iii) acceleration of sliding block A along with slotted arm CD by using Vector or space method. Here MM is your Birth Month. 12

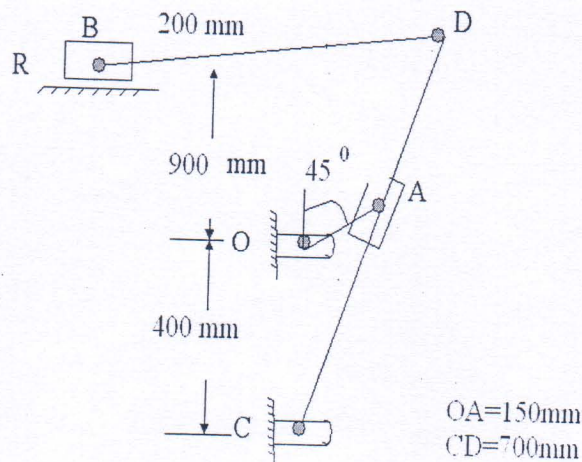


Fig. 1

- Q3 (a)** Explain cam curvature and undercutting in roller followers. **04**
- (b)** Determine displacement, velocity, acceleration and jerk for 4-5-6-7 polynomial cam in which follower rises 25mm in 600 cam rotation, when cam has turned through 250. Assume cam shaft speed to be 10rad/sec. Also derive the formula used. **07**

SECTION - II

- Q4** Determine the static equilibrium torque T_2 on crank CB in Q.2 by analytical method if the force $F = 100\text{N}$ is acting perpendicular to connecting rod BA at G. **12**

OR

- Q4 (a)** Explain the consideration of sliding friction in static force analysis with suitable example. **02**
- Q4 (b)** In Quick return mechanism shown in Q.2 figure 1. the force 1 KN acting perpendicular to DB at its mid point in inward direction. Determine the static equilibrium torque on link OA by virtual work method. **10**
- Q5** Design and draw a slider crank mechanism in which displacement of slider S is given by following relation **12**

$$S_j = \frac{66825 + 630\theta_j - 7\theta_j^2}{8100}$$

OR

- Q5** Synthesis a four bar mechanism for the following position of input and output link. **12**

$$\theta = 50^\circ, \phi = 107.2^\circ$$

$$\omega_2 = 12\text{rad/sec}, \omega_4 = 0.2325\text{rad/sec}$$

$$\alpha_2 = 0\text{rad/sec}^2, \alpha_4 = 116.5\text{rad/sec}^2$$

- Q6 (a)** Explain the Kutzbach's criteria for spatial mechanism. **05**
- (b)** Explain the analytical method for Dynamic Force Analysis. **06**